LAB ASSIGNMENT - 20.2

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TASK 1 – PASSWORD SECURITY ENHANCEMENT

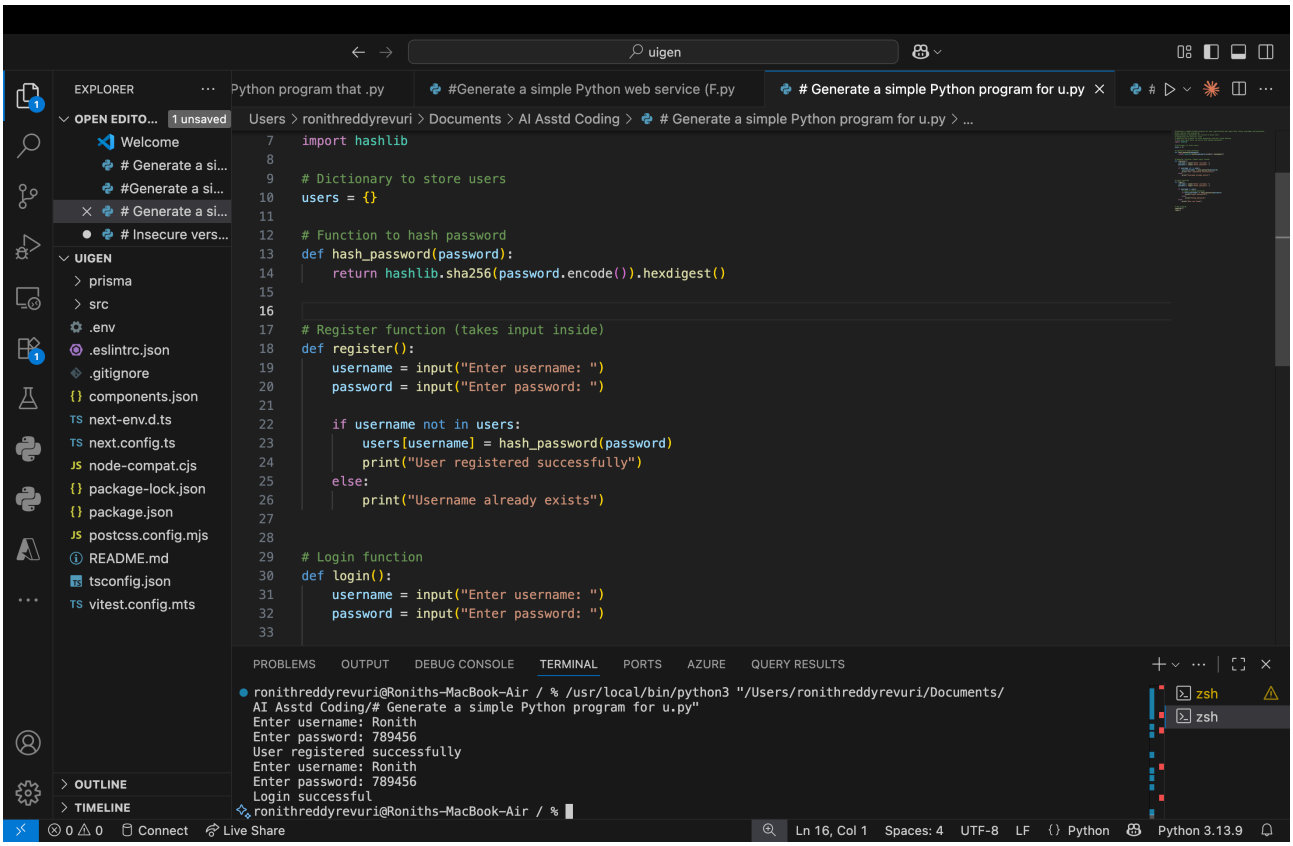
TASK:
ANALYZE AN AI-GENERATED PYTHON PROGRAM THAT STORES PASSWORDS IN PLAIN TEXT AND IMPROVE ITS SECURITY.

PROMPT :
#GENERATE A SIMPLE PYTHON PROGRAM THAT PERFORMS ENCRYPTION AND DECRYPTION USING A BASIC METHOD.
#THEN IMPROVE THE PROGRAM BY:
#IDENTIFYING WEAKNESSES IN THE ENCRYPTION METHOD
#REPLACING IT WITH A STRONGER ENCRYPTION TECHNIQUE
#ENSURING THAT DATA CANNOT BE EASILY REVERSED WITHOUT THE CORRECT KEY

EXPLANATION :

PASSWORDS ARE DIRECTLY STORED AS PLAIN TEXT IN THE FIRST VERSION, WHICH IS EXTREMELY DANGEROUS.
ALL USER PASSWORDS ARE READILY VISIBLE TO ANYONE WHO MANAGES TO ACCESS THE SYSTEM.

THE IMPROVED VERSION MAKES USE OF HASHING TO ADDRESS THIS ISSUE.



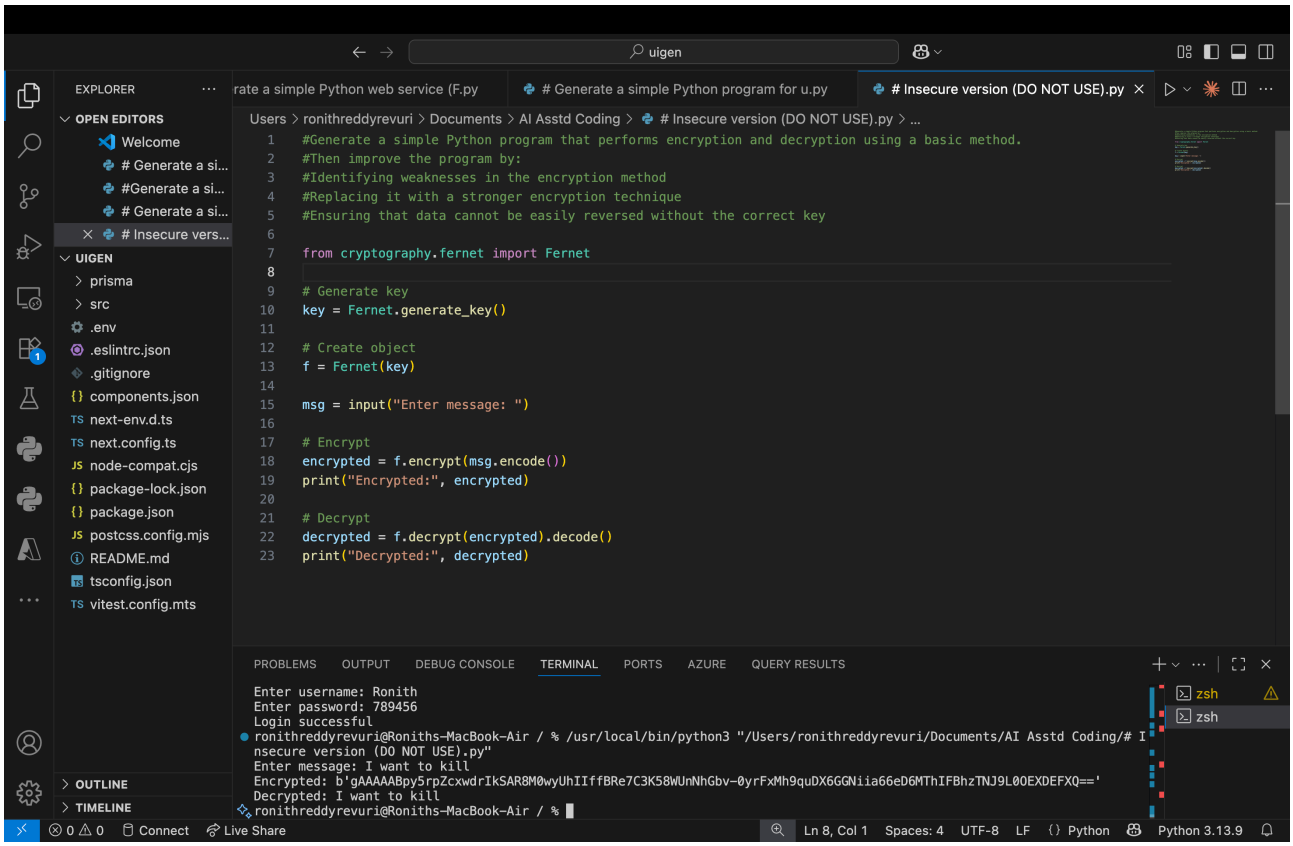
TASK 2 : SECURE API KEY MANAGEMENT
TASK:
EVALUATE AN AI-GENERATED SCRIPT THAT USES HARDCODED API KEYS AND IMPROVE ITS SECURITY.

TASK 3 : WEAK ENCRYPTION DETECTION
TASK:
EXAMINE AN AI-GENERATED PROGRAM THAT USES WEAK ENCRYPTION TECHNIQUES AND STRENGTHEN IT.

PROMPT :
#GENERATE A SIMPLE PYTHON PROGRAM THAT PERFORMS ENCRYPTION AND DECRYPTION USING A BASIC METHOD.
#THEN IMPROVE THE PROGRAM BY:
#IDENTIFYING WEAKNESSES IN THE ENCRYPTION METHOD
#REPLACING IT WITH A STRONGER ENCRYPTION TECHNIQUE
#ENSURING THAT DATA CANNOT BE EASILY REVERSED WITHOUT THE CORRECT KEY

EXPLANATION :
THE PROGRAM MAKES USE OF THE FERNET ENCRYPTION TECHNIQUE FROM THE CRYPTOGRAPHY LIBRARY TO INCREASE SECURITY.
USING A SECRET KEY, THIS TECHNIQUE ENCRYPTS AND DECRYPTS DATA.

WITHOUT THE RIGHT KEY, THE ENCRYPTED DATA CANNOT BE READ OR DECRYPTED.
WHEN COMPARED TO SIMPLE ENCODING, THIS GREATLY INCREASES SYSTEM SECURITY.

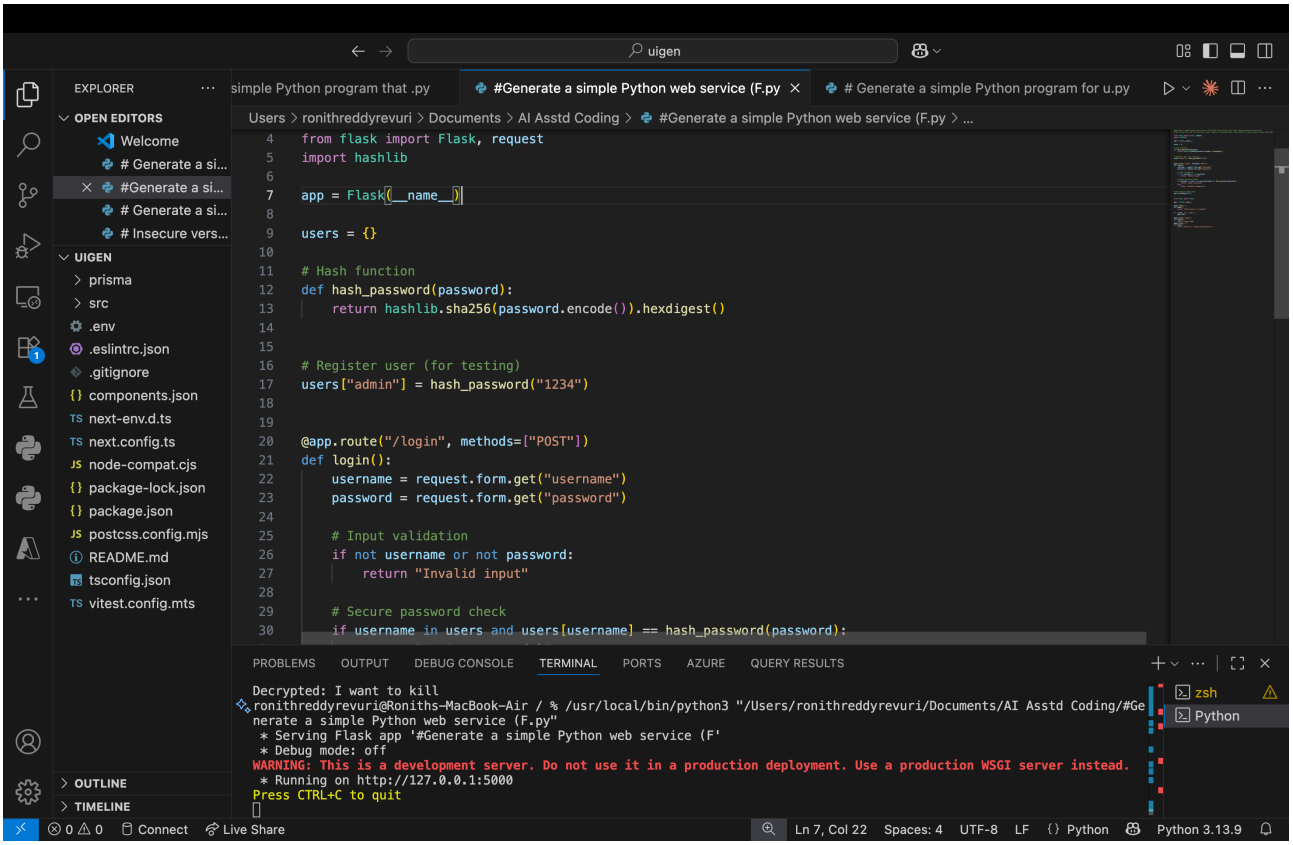


TASK 4 : REAL-TIME APPLICATION: SECURITY REVIEW OF AI-GENERATED WEB SERVICE
SCENARIO:
STUDENTS SELECT AN AI-GENERATED WEB SERVICE SNIPPET (E.G., REST API, FILE UPLOAD HANDLER, OR AUTHENTICATION MODULE).

PROMPT :
#GENERATE A SIMPLE PYTHON WEB SERVICE (FLASK API) THAT ALLOWS USER LOGIN USING USERNAME AND PASSWORD.
#THEN REVIEW THE CODE FOR SECURITY ISSUES, IDENTIFY VULNERABILITIES, AND IMPROVE IT USING SECURE CODING PRACTICES.

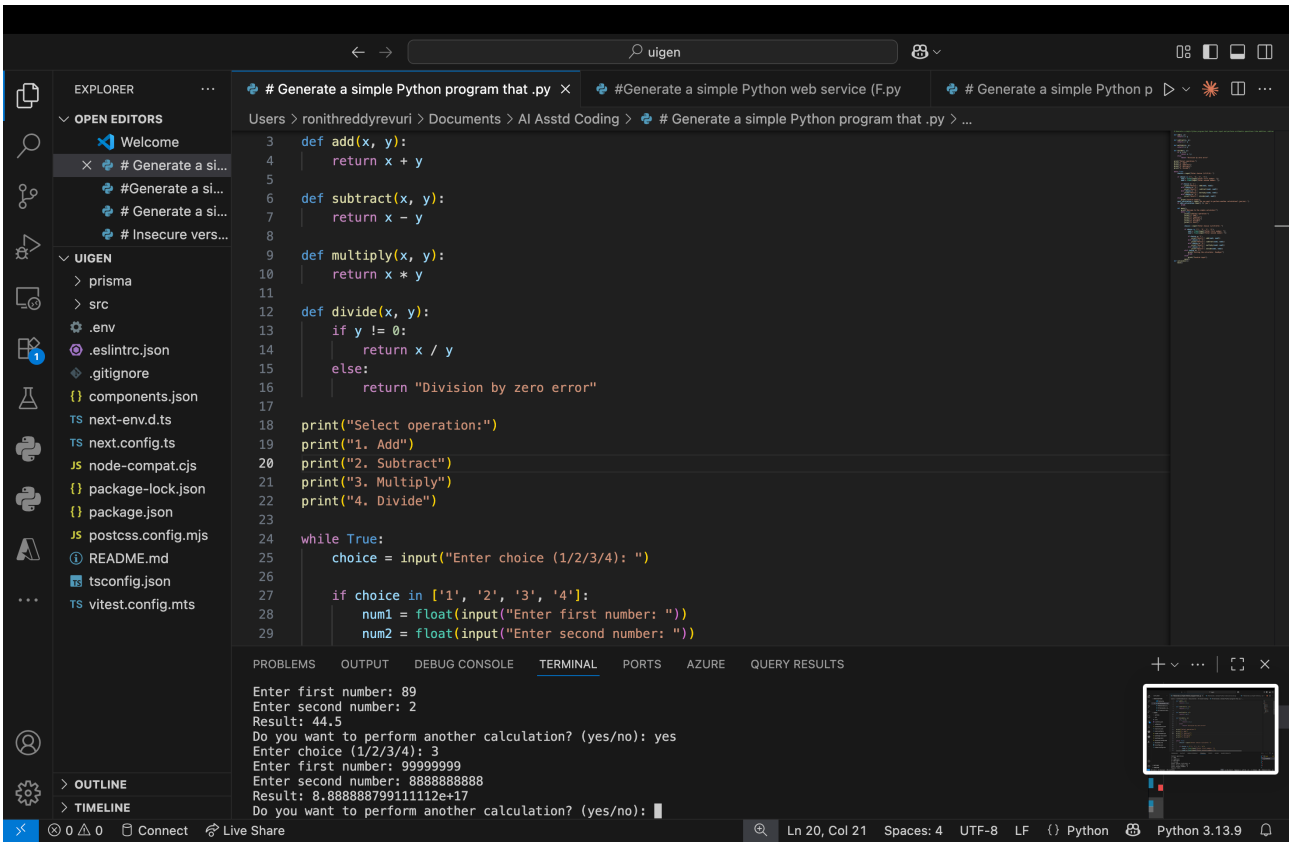
EXPLANATION :
PASSWORDS IN THE INSECURE VERSION ARE EASILY STOLEN BY ATTACKERS BECAUSE THEY ARE KEPT IN PLAIN TEXT.
ADDITIONALLY, DEBUG MODE IS ENABLED, WHICH MAY REVEAL PRIVATE SYSTEM DATA, AND USER INPUT IS NOT VERIFIED.

BEFORE BEING STORED, PASSWORDS IN THE UPDATED VERSION ARE TRANSFORMED INTO A HASHED FORMAT.



TASK 5 : EXCEPTION HANDLING IMPROVEMENT
TASK:
ANALYZE AN AI-GENERATED PYTHON PROGRAM THAT DOES NOT HANDLE RUNTIME ERRORS AND IMPROVE ITS RELIABILITY.

PROMPT :
 # GENERATE A SIMPLE PYTHON PROGRAM THAT TAKES USER INPUT AND PERFORMS ARITHMETIC OPERATIONS LIKE ADDITION, SUBTRACTION, MULTIPLICATION, AND DIVISION.

EXPLANATION :

RUNTIME ERRORS ARE NOT HANDLED BY THE ORIGINAL AI-GENERATED PROGRAM, WHICH USES USER INPUT TO CARRY OUT ARITHMETIC OPERATIONS. WHEN INCORRECT INPUT IS ENTERED, THIS COULD LEAD TO THE PROGRAM CRASHING. TRY-EXCEPT BLOCKS ARE USED TO ADD EXCEPTION HANDLING TO INCREASE RELIABILITY.